

Pitch types are relative

Can MLB's pitch labels be recovered for pitchers a model has never seen, and when the model disagrees, is the model wrong or the label?

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Code: github.com/Andresperez397/pitch-type-classifier Live app: pitch-types-are-relative.streamlit.app

DATA

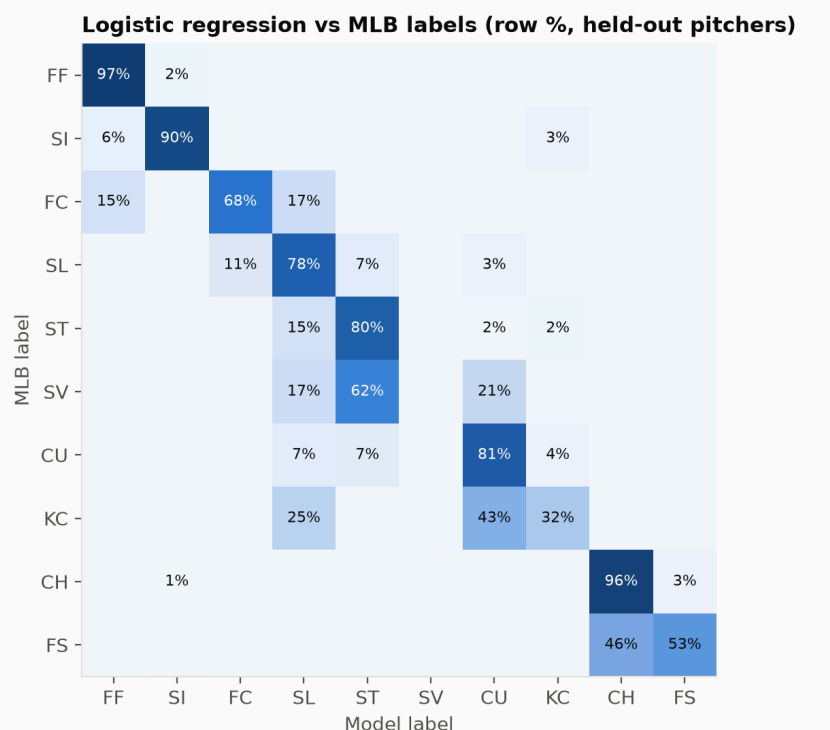
The full 2025 MLB regular season from public Statcast: 712,528 pitches from all 2,430 games, verified against MLB's official schedule. 699,305 pitches from 723 pitchers remain after cleaning. The project rebuilds a 2021 class project with stricter validation.

APPROACH

- **Pre-registered:** labels, features, models and decision rules were committed to git before any model was fit.
- **Unseen pitchers:** 20% of pitchers held out entirely, 5 repeats.
- **Pitcher-relative features:** each pitch is described against the pitcher's own hardest pitches, without using labels.
- **Error audit:** each miss is compared with what other pitchers' nearby pitches are called.

FINDINGS

- **85% accuracy on new pitchers** (macro-F1 0.66; always guessing four-seam: 32%) with a plain logistic regression. Fastballs and changeups work well; knuckle curves and slurve mostly don't.
- **Relative features helped logistic regression in 5 of 5 repeats.** For boosted trees it was 4 of 5, which doesn't meet the pre-declared rule.
- **Flexible models didn't transfer:** boosted trees trailed logistic regression even when tuned on held-out pitchers.
- **It holds in the next season:** the 2025 model scored 89% accuracy (macro-F1 0.68) on 207 pitchers who never appeared in 2025, using the complete 2026 season.
- **About two-thirds of the best model's misses sit where other pitchers' pitches carry the model's label.** The name depends on who throws the pitch.



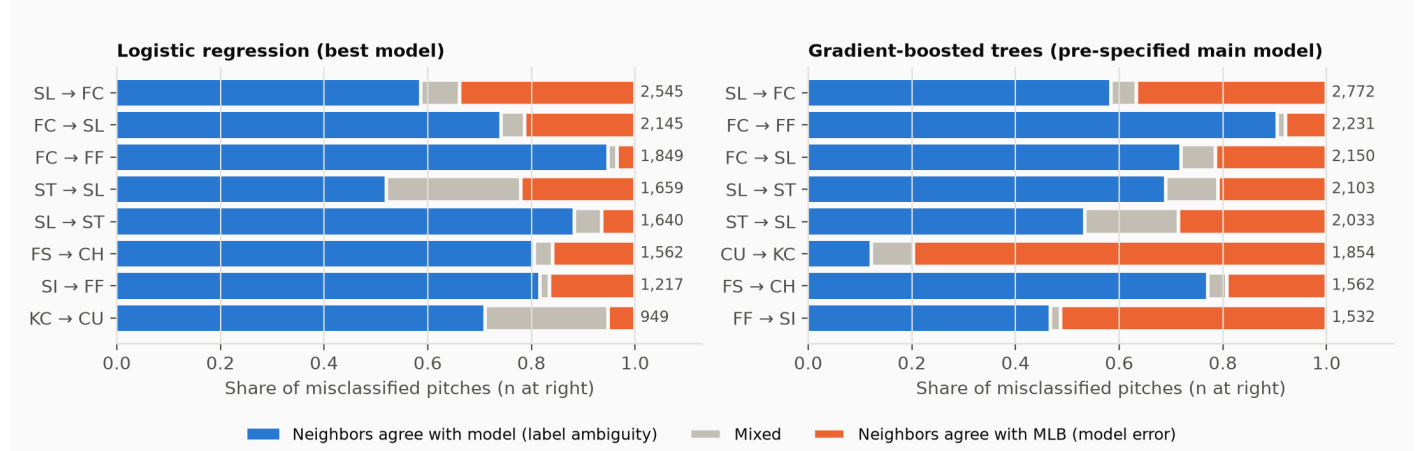
Row-normalized confusion matrix for held-out pitchers (best model). Most confusion sits at naming boundaries: cutter/four-seam, slider/sweeper/cutter, splitter/changeup, knuckle curve/curveball.

MODEL COMPARISON (RELATIVE FEATURES, HELD-OUT PITCHERS)

Model	Accuracy	Macro-F1	Log loss
Logistic regression	0.849	0.660	0.43
Boosted trees, plan's fixed settings	0.807	0.619	2.86
Boosted trees, tuned on held-out pitchers (exploratory)	0.840	0.649	—

Means over 5 repeats; each repeat holds out a different 20% of pitchers.

Most common disagreements: model error or label ambiguity?



For each misclassified pitch: do most of its 50 nearest pitches from other pitchers carry MLB's label (model error) or the model's label (label ambiguity)?

INSTRUCTIVE CASES

- **Kenley Jansen:** his signature "cutter" is his fastball, and the model reads it as one.
- **Tyler Rogers (submarine):** his 83 mph sinker drops like a curveball. A population model struggles at extreme arm angles.

WHY IT MATTERS

Pitch labels are naming decisions, not physical categories. Models that use labels across pitchers (pitch-type splits, pitch-quality models) inherit that inconsistency. Pitcher-relative descriptions are safer.

LIMITATIONS

MLB's labels are the target, not ground truth. The analysis covers one season and excludes pitchers with fewer than 100 pitches. The hardest-pitch reference assumes the fastest pitches are fastballs. Slurves and knuckle curves are rare classes.

Data: Baseball Savant / Statcast (MLB Advanced Media), public pitch-level data. Interactive arsenal explorer: pitch-types-are-relative.streamlit.app.